Krishna 2021 ccRCC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

UT2_Center + UT2_Far intermix on one continuous manifold; 0 detached islands, single Leiden compartment (θ=10, mixing plateau). Judged on the embedding, not a scalar alone.

Gate 2 — module geography → GO (non-branching)

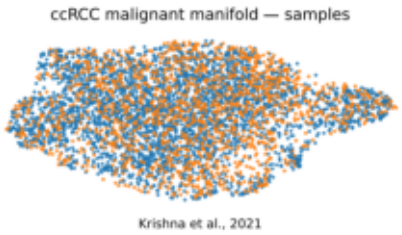
Non-branching: a single pSom→Crest gradient (pSom_sm–Crest_sm r=−0.89); Crest terminal (crest-high decile mean pt=0.97); Gast weak/diffuse.

Gate 3 — cross-repression → NA

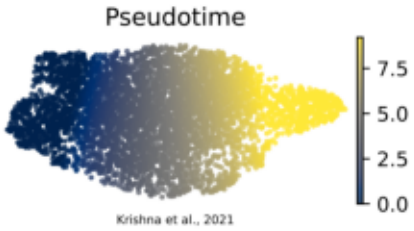
No bifurcation (2 tips, 0 forks) — the competitive Gast↔pSom arm test does not apply. Non-branching case, as in prior Kidney runs (Bi/Zhang).

STEP FIGURES

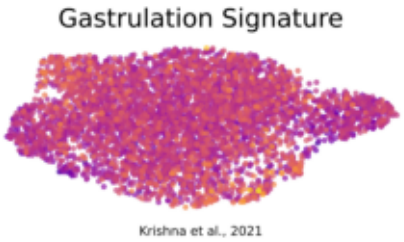
Manifold — samples



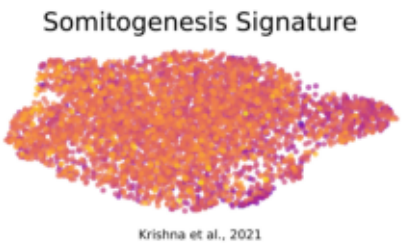
Pseudotime



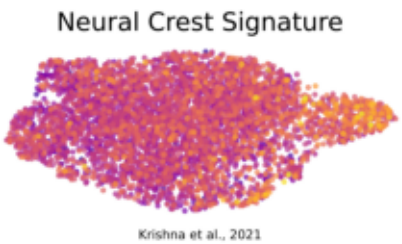
Gastrulation



Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Clear-cell RCC (Krishna et al., 2021), 10x; 5,715 malignant cells pre-filter, 5,006 after thresholding (all CD45− ccRCC CA9+). One continuous non-branching manifold.

Verdict.
Pattern reproduces (Linear-pSom/Crest). A single embryonic axis rooted at the somitogenesis-high origin flows to a neural-crest terminus (pSom ρ=−0.85, Crest ρ=+0.75 vs pseudotime); no Gast/pSom bifurcation, so Gate 3 is NA. Caveat: the clean 15.9× count gap retains only two regions of one patient (UT2), 5,006/5,715 cells, so integration is region-level; cycling ρ=−0.23 vs pt (not proliferation-driven).

Parameter choices

Min cells/sample	403 (geom-mean 15.9× gap; keeps UT2)	**Integration**	Harmony θ=10; PCA20; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip 2 = Gast/pSom-high, Crest-low	**Geometry**	Linear (non-branching); 2 tips / 0 forks
EMT-TFs	TWIST2 (early) · ZEB2 (late)		